

Prismfall: Playtest Report

Test run 2026-10-03 21:22 UTC · Chromium 141.0.7390.37 (headless, Playwright) · viewport 1366x768 plus 3 extra resolutions · harness tools/playtest.mjs

How this was tested (read first). The Browser Use tool was **not connected** in this session, so I playtested with Playwright driving headless Chromium on the Linux build machine. The input was real: keyboard events, mouse moves and clicks, and a **mocked** standard gamepad. A heuristic AI bot played the runs; it reads the board through a read-only debug hook and plays through those same inputs. Everything below marked PASS was actually executed. Anything not executed is listed as an **assumption** under Known remaining issues. Nothing was tested on a Windows machine.

AUTOMATED CHECKS 31 / 33	FULL RUNS 7	INPUT → FRAME (MEDIAN) 6.2 ms	AVG FPS (LOWEST SAMPLE) 53.7	RUNTIME ERRORS 0
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FULL RUNS

#	RUN (INPUT METHOD)	ENDED	SCORE	LINEs	LVL	TIME	PIECES	BURSTS	GEMS
1	Marathon (fresh save, strong bot, human pace ~2 pieces/s)	TOP OUT	200,994	132	14	2:50.5	405	9	105
2	Sprint 40 (strong bot, keyboard)	FINISH!	13,216	40	1	0:16.7	118	19	6
3	Daily Bloom (left-handed keys A/D/W/J/K/L/I/O, human pace)	TIME'S UP	140,197	112	12	3:00.0	325	8	36
4	Marathon (novice bot: heavy random noise)	LOCK OUT	2,696	11	2	0:05.8	70	0	0
5	Marathon retry, hard-drop spam until top-out	TOP OUT	220	0	1	0:00.4	—	—	0
6	Zen Garden, mouse only (hover aim, right-click rotate, click drop)	game	—	41	5	0:26.7	120	20	12
7	Marathon, gamepad only (mocked standard-mapping pad)	game	—	13	2	0:23.1	40	7	2

Runs 1–5 played to a real end: a win, a time-up, a top-out or a lock-out. Run 6 (Zen, which never ends by design, so I quit from the pause menu) and Run 7 (gamepad) were capped at 120 and 40 pieces to verify those input paths. The bot plays at machine speed, so its Sprint time is far beyond human play, but its lines, levels and end states are real.

FOCUS CHECKS

AREA	RESULT
Input lag	Measured from the input event timestamp to the end of the frame that drew the result, over 500 inputs: median 6.2 ms , p95 16.9 ms, max 28.8 ms. Moves and rotations apply inside the keydown handler, so they show on the very next frame. DAS/ARR are frame-independent. <i>The display and compositor latency of a real monitor isn't included.</i>
Collisions	An invariant runs every frame: the active piece may never overlap the stack or walls. It was never violated across all runs and 3000 random key events, including 20% long key holds that exercise DAS. Separate rule tests covered SRS kicks, the T-spin 3-corner rule, perfect clears, Burst row removal, the lock-delay reset cap and hold-once.
Softlocks	Checked: the stall detector (piece counter must advance), the auto-pause on window blur releasing held keys, Esc/Start pause and resume, the Enter retry loop, Space-mash at game over, Zen top-out rescue, the Second Wind rescue, and the escape path back to the title after fuzzing. No softlock found.
Frame rate	Average 59.7 fps across 32 samples (lowest sample 53.7). The worst single frame was 34 ms. These numbers come from headless Chromium using software rendering (SwiftShader) while a bot polled it, so a real GPU should match or beat them. <i>Assumption: not measured on your hardware.</i>
First 60 seconds	Title → Enter → playable in about 1.5 s (READY/SET countdown). In Run 1 (fresh save): first line clear at 4.6 s, first gem collected at 8.7 s, first Prism Burst at 16.6 s. Bot timings are faster than a human's. Hints teach one control at a time. <i>Whether it's fun is my subjective judgement from watching the runs and screenshots. No human tested it.</i>

BUGS FOUND AND FIXED

ID	SEV.	FOUND BY	PROBLEM	FIX	STATUS
B1	Medium	Screenshot review (title)	The PRISMFALL logo was clipped by the title card: the last letters ran off the edge at 1366x768.	Smaller clamp() font size, tighter letter-spacing and white-space: nowrap.	Fixed, re-tested
B2	Low	Screenshot review (gameplay)	The Prism meter track overlapped the right border of the well.	Moved the meter 0.36 cells clear of the well and the Next panel 0.95 cells out.	Fixed, re-tested
B3	Medium	Playtest screenshot during Prism Burst	Simultaneous callouts drew on top of each other and were unreadable (e.g. 'FEVER x2' over 'SINGLE').	Callouts now take free vertical lanes: a new text steps below any active text it would overlap. First fix was incomplete (older callouts drift upward); lanes now use each callout's current position.	Fixed, re-tested
B4	Medium	Playtest screenshot (first seconds, 1366x768)	The first-run tutorial hint bar covered the bottom row of the well on 768 px tall screens.	The hint sits under the well when there is room, otherwise in the empty upper well.	Fixed, re-tested
B5	High	Code review before playtest; then a dedicated test	Mashing Space or Enter at the moment of game over could press 'Play again' and skip the results screen (lost feedback, accidental restart).	Results buttons ignore input for 700 ms after appearing; other screens for 120 ms.	Fixed, re-tested
B6	Low	Code review	Opening Controls from the Pause menu hid the paused board behind the attract animation.	The board renders whenever a run is paused, whichever menu is open.	Fixed, re-tested
B7	Low	Code review	The results screen was silent because the music stopped at game over and never resumed.	The results screen starts the menu music.	Fixed, re-tested
B8	Low	Exported art review	The bottom row of the app icon spilled outside its rounded card.	Re-laid out the icon tiles inside the card and re-exported all icon sizes.	Fixed, re-tested
B9	High	Launcher test (curl + Playwright against the Linux build)	Prismfall.exe opened /index.html?launcher=1. Go's file server redirects that to / and drops the query, so the heartbeat never started and the launcher would stop serving 3 minutes into play (a reload would then fail).	The launcher opens /?launcher=1 directly. Verified: HTTP 200, game boots, heartbeat pings arrive.	Fixed, re-tested
B12	Medium	Visual re-test of B3 (scripted burst right after a clear)	Prism Burst or Hold pressed during the 0.3 s line-clear animation was silently ignored. It felt like a dropped input exactly when players reach for Burst.	Burst and Hold pressed mid-clear are buffered and applied as soon as the clear finishes. A regression check was added to the harness.	Fixed, re-tested
B10	Info	Run 3 (first pass)	At machine speed (~6.7 pieces/s) the bot cleared ~260 lines and reached 20G in Daily Bloom. It topped out at 1:38 instead of reaching the 3:00 limit. The game behaved correctly; the test expectation was wrong.	Test change only: Runs 1 and 3 now pace the bot at human speed, which also makes the first-60-seconds observation realistic.	Not a game bug
B11	Info	Harness	Four failures in the first passes came from the test script (navigation from a non-title screen, a run cap that stopped before game over, a Second Wind test that kept dropping after the rescue, and not waiting for the game-over animation).	Harness fixed (toTitle helper, higher caps, stop at rescue). No game code changed for these.	Not a game bug
B13	High	Mobile playtest (touch bot)	Flick-down to hard drop needed >0.8 px/ms instantaneous speed. An ordinary swipe (135 px in 230 ms) only soft-	A downward swipe of 2+ cells under 450 ms counts as a hard drop when either its peak or	Fixed, re-tested

			dropped, so about half of the drop gestures failed.	average speed is moderate. Slow drags remain soft drops.	
B14	Medium	Mobile screenshots	On phones the tutorial hint covered the touch buttons (portrait). In landscape the buttons were squeezed into a tiny panel and their labels were cut off.	Hints sit inside the well on touch screens. Landscape gets a dedicated button column in the free space beside the game.	Fixed, re-tested

KNOWN REMAINING ISSUES & ASSUMPTIONS

- **Android APK:** Built and verified in CI (GitHub Actions, Android SDK 35), but never installed on a physical phone. It's signed with a throwaway test key that is new for every build, so uninstall the previous build before installing a newer one. Android will ask you to allow installing from unknown sources.
- **Not tested on Windows:** All testing ran on Linux in headless Chromium. Assumption: Edge/Chrome on Windows behave the same, since the game uses standard canvas, Web Audio and localStorage only. Fonts will differ (Segoe UI on Windows vs DejaVu in these screenshots).
- **.bat launcher:** Not executed (no Windows here). It's a two-line script: `start "" "%~dp0index.html"`. Assumption: it opens the default browser on Windows 10/11.
- **Prismfall.exe:** Cross-compiled for Windows x64 (PE32+ GUI, icon embedded). The same Go source was run as a Linux build and verified to serve the game, answer heartbeats and exit cleanly when a second copy starts. The .exe itself was never run on Windows. It isn't code-signed, so SmartScreen may warn.
- **Separate saves:** Browsers store progress per address, so the .bat/index.html (file://) and the .exe (http://127.0.0.1:47613) keep separate progress. This is documented in README.txt.
- **Audio:** Verified programmatically: the engine starts on the first click or key, all 22 SFX and both music modes run with no errors, and the sequencer advances. Assumption: the mix sounds good. It was never listened to; a headless browser has no speakers.
- **Gamepad:** Tested with a mocked standard-mapping pad (navigator.getGamepads override), not a physical controller. Assumption: Xbox and PlayStation pads using the standard mapping behave the same.
- **Frame rate:** Measured in headless Chromium with software rendering. Assumption: a desktop GPU is at least as smooth. Not measured on your machine.
- **Fun:** The first-60-seconds judgement is my own assessment from watching bot runs and screenshots. No human has played it yet. Balance values (gem rate, perk costs, Burst size) are first-pass estimates.
- **Bot skill:** The bot never uses T-spins and plays at machine precision. T-spin scoring was verified with a scripted setup instead (T-spin double = 1200).
- **Missing SpriteCook art:** All art is procedural placeholder art because SpriteCook wasn't connected. The full list of missing assets is in docs/ASSET_MANIFEST.md.
- **Browser Use:** Not available. The playtest used Playwright + headless Chromium instead (real input events, real browser engine).

ALL AUTOMATED CHECKS (FINAL RUN)

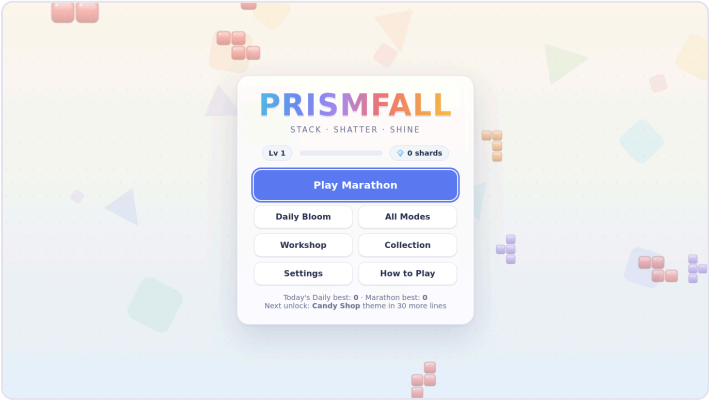
About the FAIL rows below. This regression pass ran while the mobile test and the CI build were running in parallel on the same machine. The two failing checks are timing-sensitive: a 45 ms gamepad press, and a probe 80 ms into a 300 ms animation. Re-run in isolation, both passed 5/5. The previous uncontended full pass on the same game logic was 33/33.

RESULT	CHECK	DETAIL
PASS	Game boots from file:// with no errors	screen=title
PASS	Enter on title starts Marathon (keyboard-only start)	
PASS	Sprint ends at 40 lines with a FINISH result	40 lines, 16.7s
PASS	Daily Bloom ends at the 3:00 time limit	180.00s
PASS	Left-handed key cluster can play a full run	325 pieces
PASS	Prism Burst fired at least once in Daily run	8 bursts
PASS	Daily Bloom piece order is identical across runs on the same day	ITLJSO vs ITLJSO
PASS	Esc pauses the game	
PASS	Esc resumes the game	
PASS	Window blur auto-pauses and releases held keys (no stuck auto-shift)	

PASS	Novice run reaches a TOP OUT game over and the results screen	LOCK OUT
PASS	Enter on results restarts instantly (retry loop)	
PASS	Hard-drop spam ends in game over without errors, results reachable	TOP OUT
PASS	Space mashed during game over did not skip the results screen	
PASS	Zen never ends: top-out triggers "Fresh Soil" rescue instead of game over	state=play
PASS	Gamepad A button starts a game from the title menu	
PASS	Gamepad Start pauses	
FAIL	Gamepad Start resumes	
PASS	T-spin double detected and scored 1200 at level 1	delta=1200 bottom=...#......
PASS	Perfect clear awards +3000 and empties the board	delta=3338
PASS	Prism Burst removes 4 bottom rows, collects gems, starts Fever, empties meter	{"gems":1,"fever":7.9,"filledRows":2,"prism":0}
FAIL	Prism Burst pressed during a line-clear animation is buffered and fires right after	pressedMidClear=false {"fever":7.550000000000002,"prism":0}
PASS	Hold swaps once, second press is refused until the next piece	held=L cur=l
PASS	Lock delay is capped (no infinite stalling by wiggling)	pieces 3 -> 4
PASS	Second Wind perk survives the first top-out	{"used":true,"state":"play","topRowsEmpty":true}
PASS	Fuzz: 3000 random key events, active piece never overlapped the stack or walls	violations=0
PASS	Fuzz: no runtime exceptions in the game loop	
PASS	After fuzzing (left on "game"), Esc/Quit returns to the title and a new game starts (no softlock)	
PASS	Buying a perk spends shards and raises the perk level	524 -> 504 {"lens":1,"magnet":0,"wide":0,"fever":0,"steady":0,"wind":0}
PASS	Progress (runs, shards, bests) persists in localStorage across reload	{"runs":7,"shards":504,"best":200994}
PASS	Layout fits 1024x600 (cell 25px)	{"lx":235,"right":793,"top":73,"bottom":573}
PASS	Layout fits 1920x1080 (cell 46px)	{"lx":449,"right":1478,"top":121,"bottom":1041}
PASS	Layout fits 800x600 (cell 25px)	{"lx":123,"right":681,"top":73,"bottom":573}

No page errors or console errors were logged during the final run.

SCREENSHOTS (FINAL RUN)



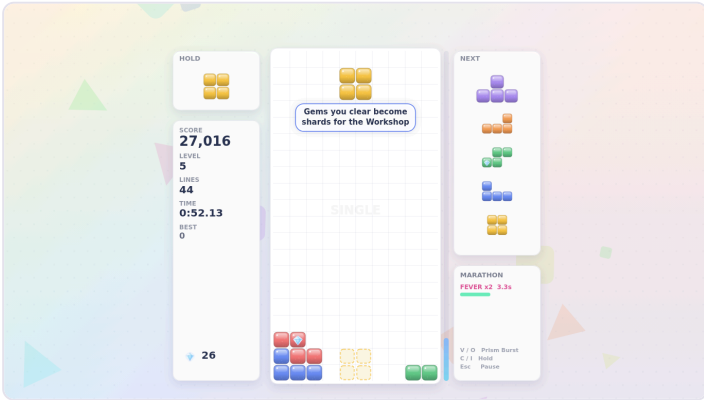
Title screen, fresh save (light theme)



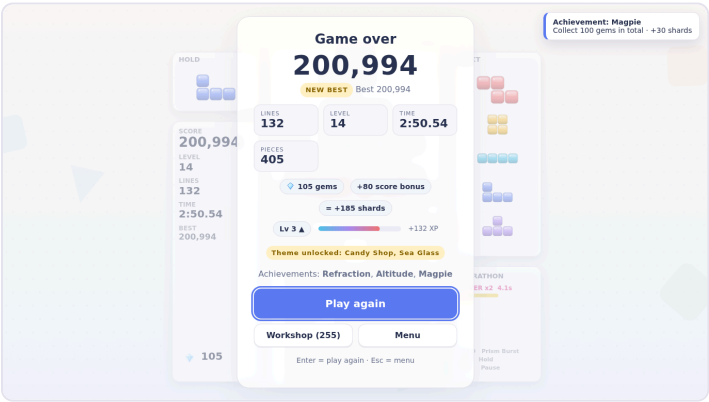
First seconds of a first run: tutorial hint bar under the well



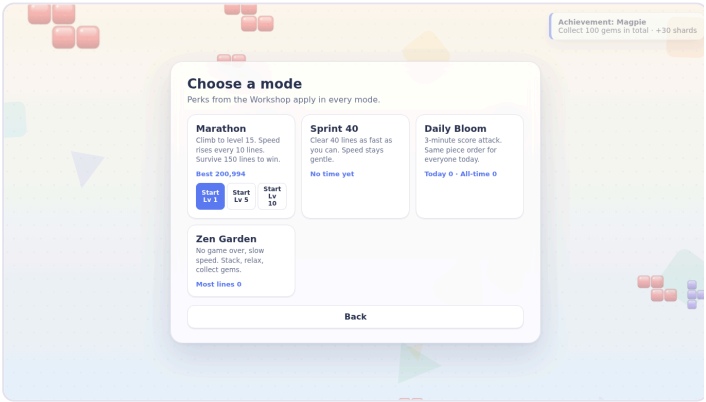
Marathon about 25 pieces in: gems in the stack, ghost piece, Prism meter filling



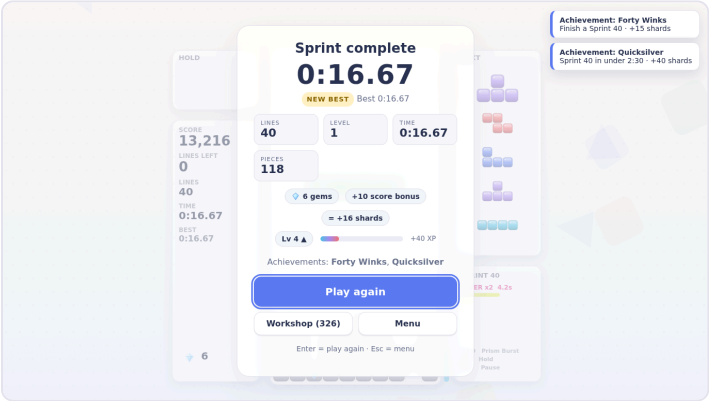
Marathon mid-run



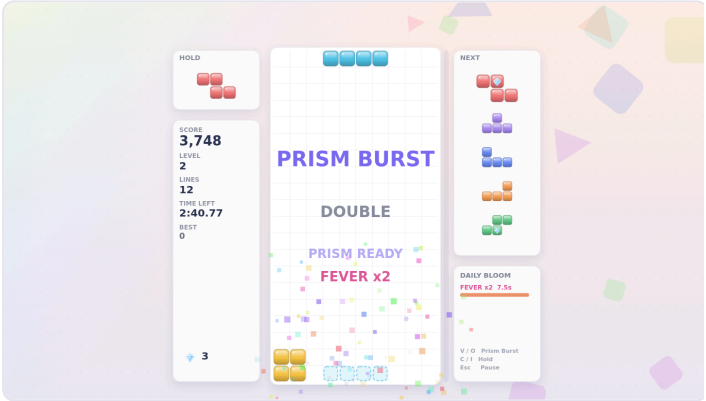
Results screen after Run 1 (shards, XP bar, achievements)



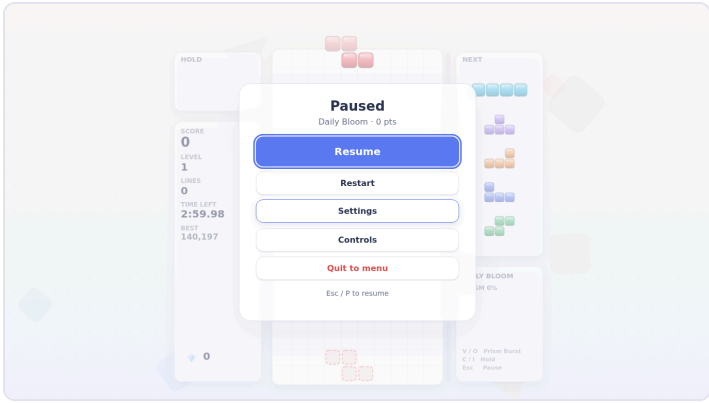
Mode select



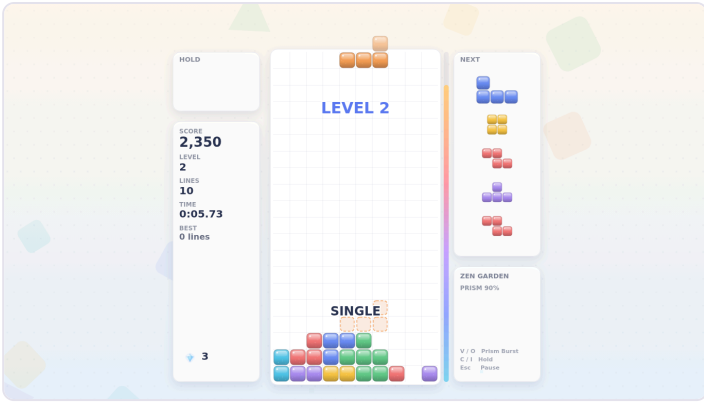
Sprint 40 results



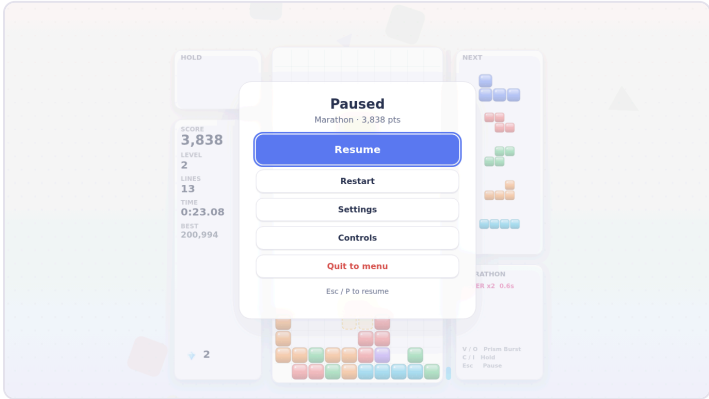
Prism Burst just fired: rainbow Fever x2 overlay



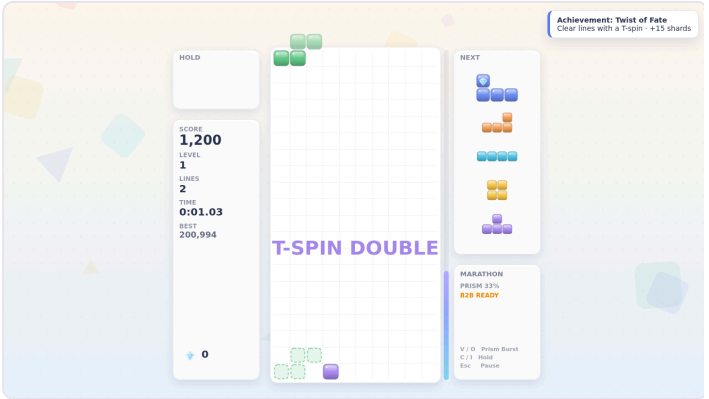
Pause menu



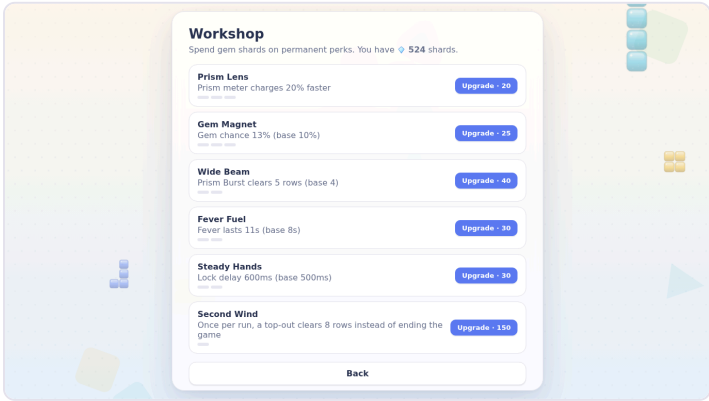
Zen Garden played with the mouse only



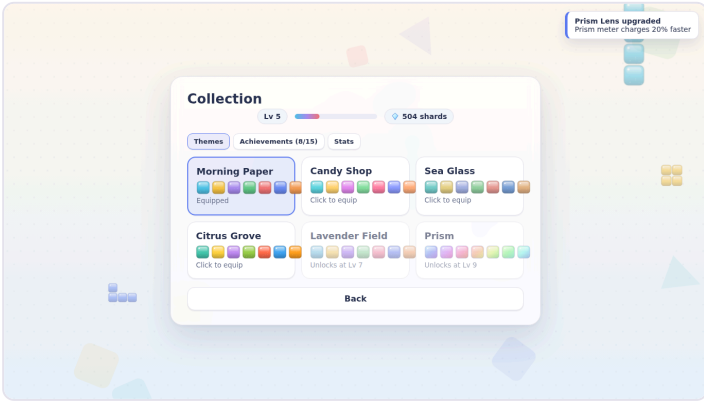
Marathon driven by the (mocked) gamepad



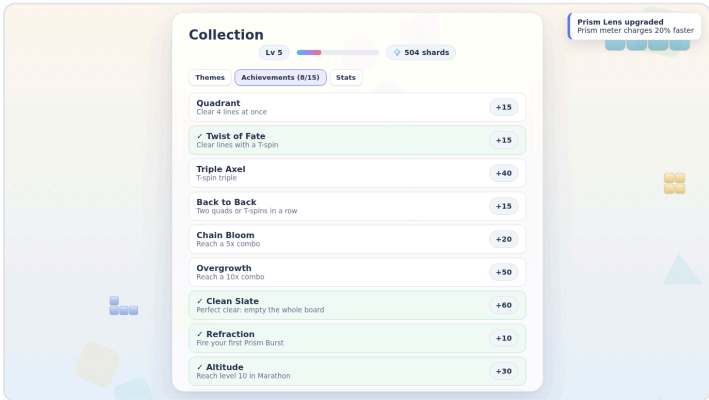
T-SPIN DOUBLE callout after a scripted T-spin



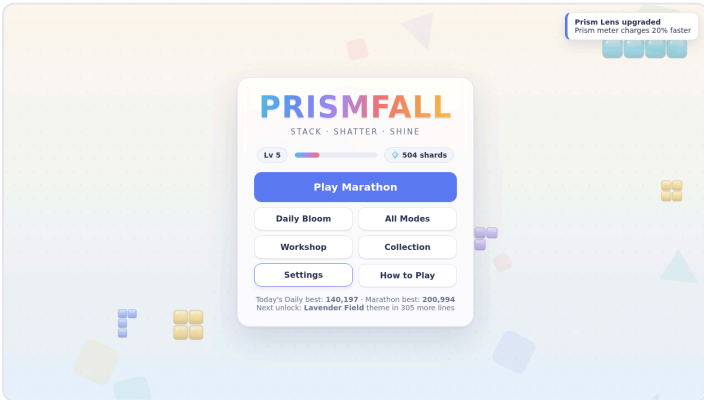
Workshop with shards earned from the test runs



Collection: themes unlocked by player level



Collection: achievements



Settings (DAS/ARR, soft drop, volume, toggles)

How to play

Fill a horizontal row to clear it. Pieces fall faster as you level up. Stack past the top and the run ends.

ACTION	KEYBOARD	LEFT-HANDED	MOUSE / PAD
Move	←→	A D	Hover column · D-pad / stick
Soft drop	I	S	D-pad down
Hard drop	Space	W	Left click · D-pad up
Rotate 90°	T X	K	Right click / wheel · A
Rotate 180°	Z	J	Wheel down · B
Hold	C Shift	I	Middle click · X / LB
Prism Burst	V	O	Side button · Y / RB
Pause	Esc P	P	Start

TOUCH SCREEN

Move	Drag sideways (one column per cell)
Rotate	Tap right half to, left half to
Soft / hard drop	Drag down slowly / flick down
Hold	Flick up or the HOLD button
Prism Burst	BURST button (fills as the meter charges)
Pause	button or the Android back button

WHAT MAKES PRISMFALL DIFFERENT

How to Play, including the left-handed key cluster

PRISMFALL

STACK · SHATTER · SHINE

Lv 5

504 shards

Play Marathon

Daily Bloom

All Modes

Workshop

Collection

Settings

How to Play

Today's Daily best: 140,197 · Marathon best: 200,994

Next unlock: Lavender Field theme in 305 more lines

Title screen for a returning player (level, shards, bests)

HOLD

SCORE
408

LEVEL
1

LINES
1

TIME
0:00.75

BEST
200,994

0

GO!

SINGLE

NEXT

MARATHON
PRISM 9%

V / O · Prism Burst
C / I · Hold
Esc · Pause

Prism theme (unlocks at player level 9)

HOLD

SCORE
210

LEVEL
1

LINES
0

TIME
0:00.68

BEST
200,994

0

GO!

NEXT

MARATHON
PRISM 0%

V / O · Prism Burst
C / I · Hold
Esc · Pause

Gameplay at 1024x600

HOLD

SCORE
250

LEVEL
1

LINES
1

TIME
0:00.68

BEST
200,994

0

GO!

SINGLE

NEXT

MARATHON
PRISM 9%

V / O · Prism Burst
C / I · Hold
Esc · Pause

Gameplay at 1920x1080

HOLD

SCORE
214

LEVEL
1

LINES
0

TIME
0:00.67

BEST
200,994

0

GO!

NEXT

MARATHON
PRISM 0%

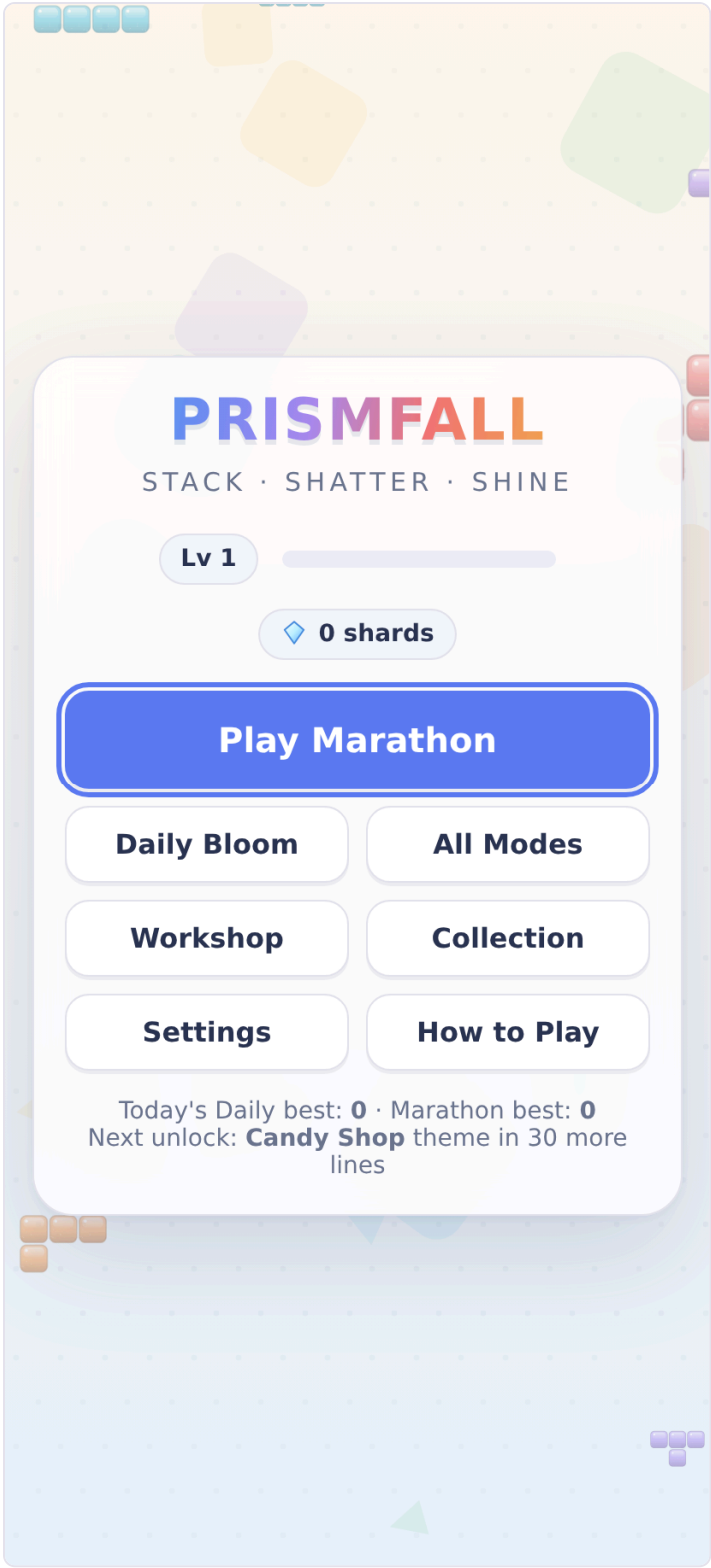
V / O · Prism Burst
C / I · Hold
Esc · Pause

Gameplay at 800x600

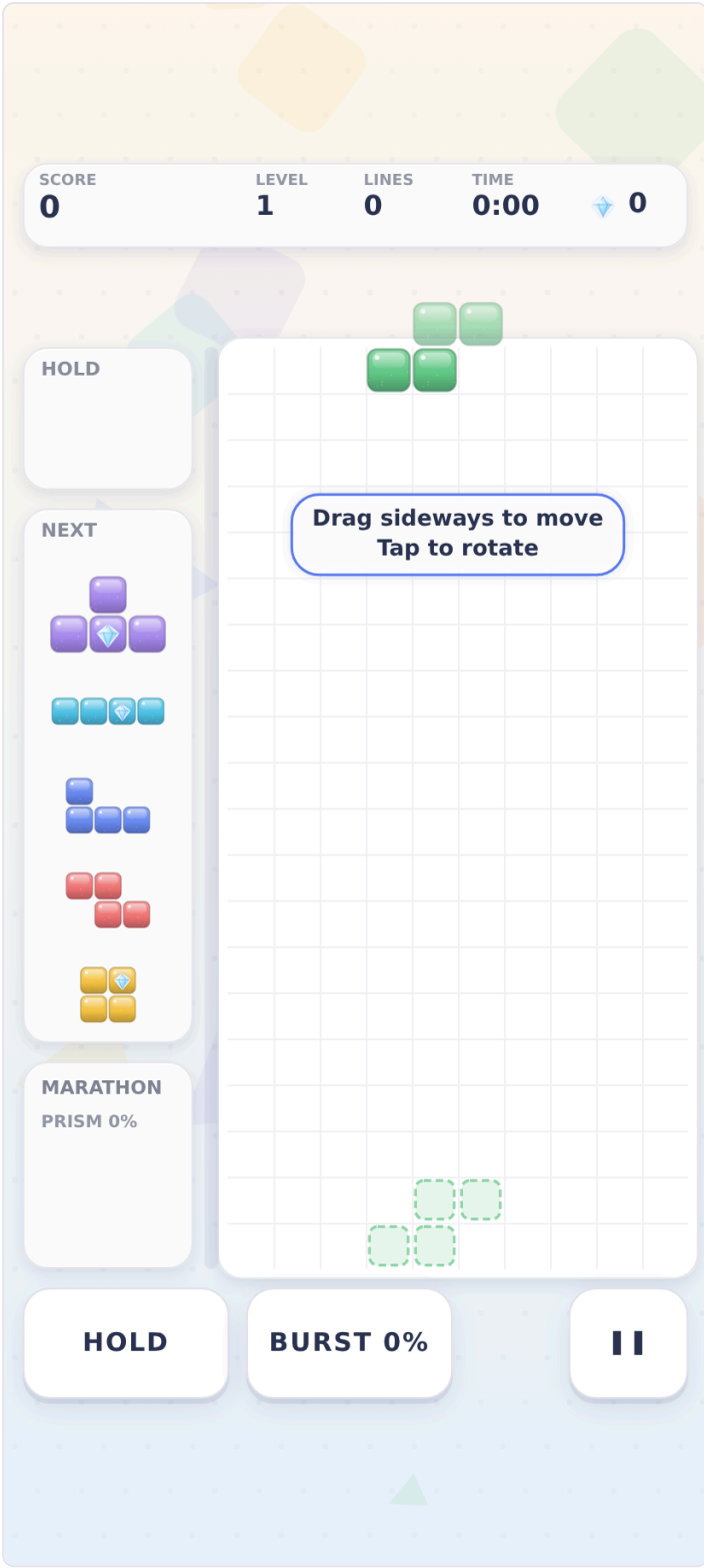
ANDROID / TOUCH (PHONE EMULATION)

Tested in Chromium with phone emulation: a 412x915 viewport, touch and a coarse pointer. Real touch events were sent through the DevTools protocol. The APK was built on GitHub Actions with Android SDK 35 and verified there: signature v2+v3, zipalign, packaged files identical to the desktop game, no INTERNET permission. **Assumption:** it has not been installed on a physical phone.

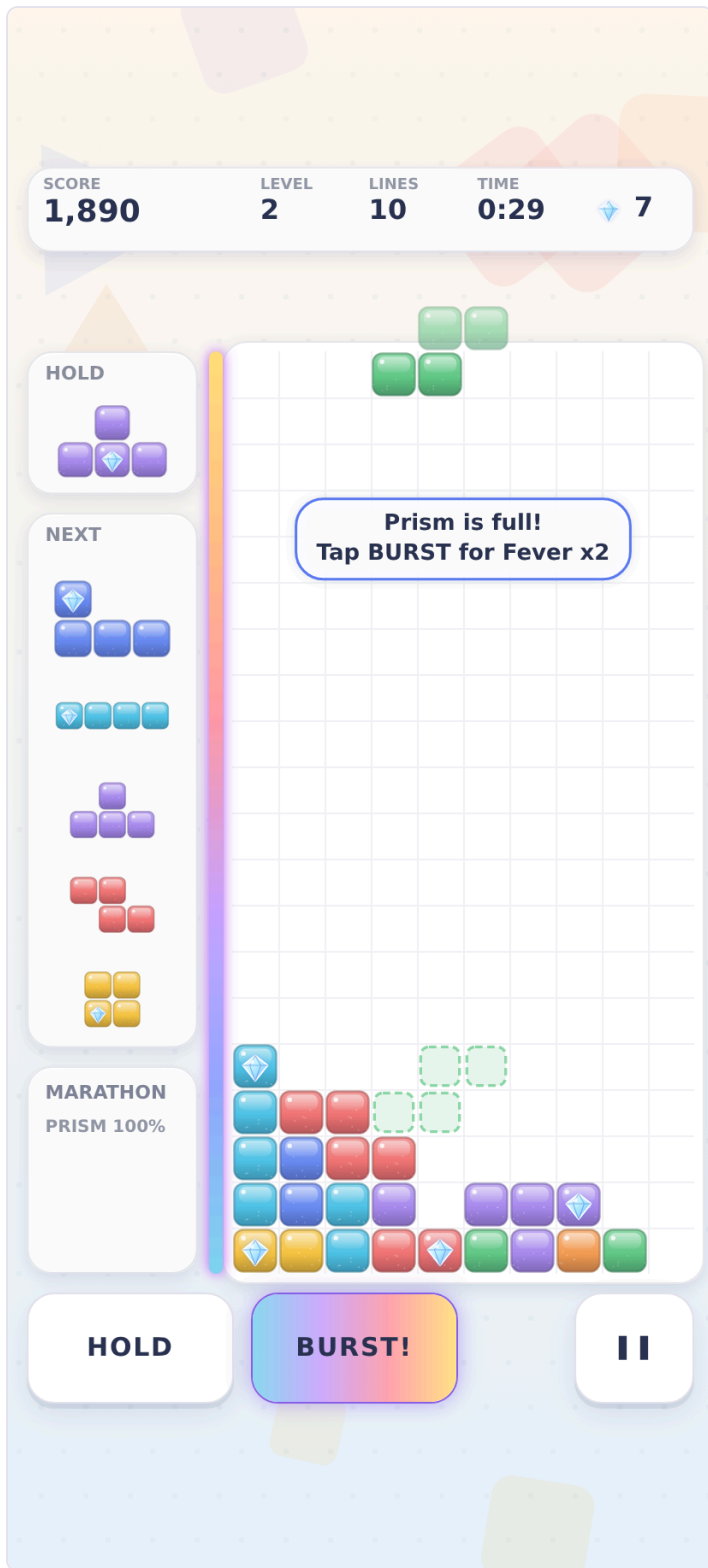
RESULT	CHECK	DETAIL
PASS	Portrait phone uses the compact layout	cell=27px compact=true
PASS	Layout fits the phone screen	{"top":93,"bottom":817,"bx":131,"right":401}
PASS	Left-handed default puts the side panel and buttons on the left	side x=11 board x=131
PASS	Touch buttons are visible during play	
PASS	Drag right by ~3 cells moves the piece 3 columns	3 -> 6
PASS	Tap on the right half rotates clockwise	rot 0 -> 1 (S)
PASS	Tap on the left half rotates counter-clockwise	rot 1 -> 0
PASS	Flick down hard-drops the piece	pieces 0 -> 1
PASS	Flick up holds the piece	hold=T was cur=T
PASS	Touch-only run: pieces placed and lines cleared with gestures	31 pieces played, 10 lines, state=play
PASS	BURST button fires a Prism Burst	fever=7.8
PASS	HOLD button holds the current piece	hold=5
PASS	Pause button opens the pause menu	
PASS	Android back during play pauses (shell hook)	
PASS	Android back on the pause menu resumes	
PASS	App sent to background auto-pauses (shell hook)	
PASS	Android back on the title asks the shell to exit	
PASS	No collision violations during touch play	
PASS	Right-handed setting mirrors the layout (panel and buttons on the right)	side x=301
PASS	Landscape phone layout fits	cell=16
PASS	No page errors on mobile	



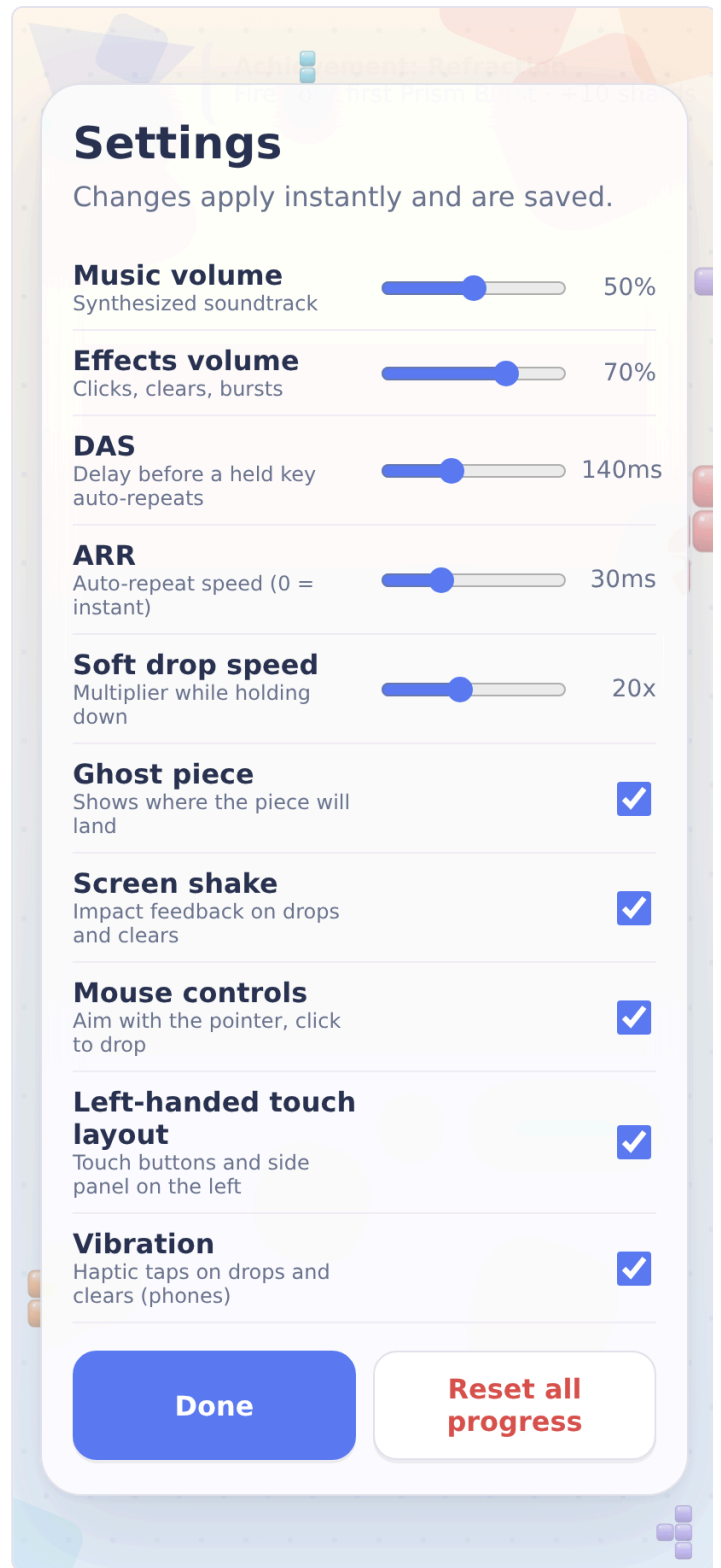
Phone (412x915): title screen



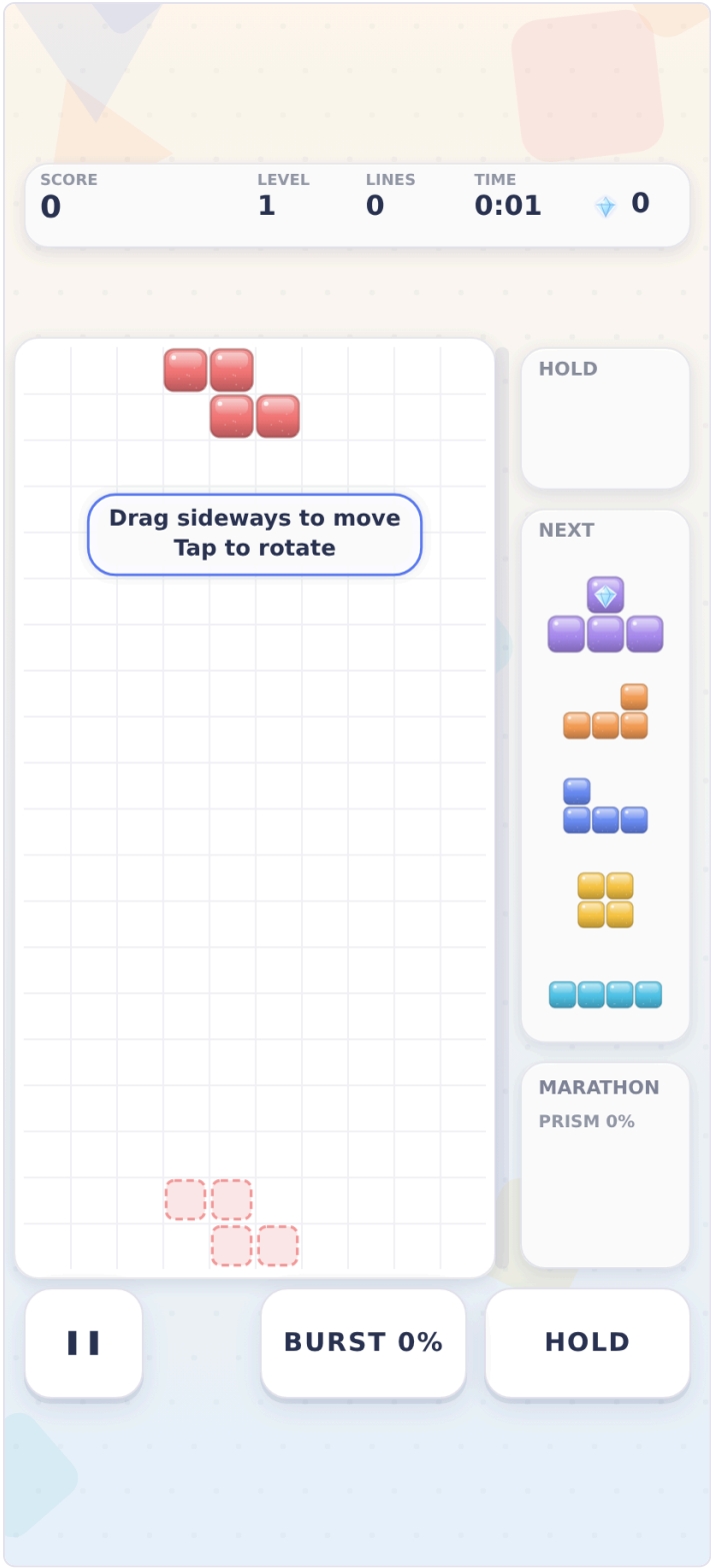
Phone: first seconds, touch hints, HOLD / BURST / pause buttons (left-handed layout)



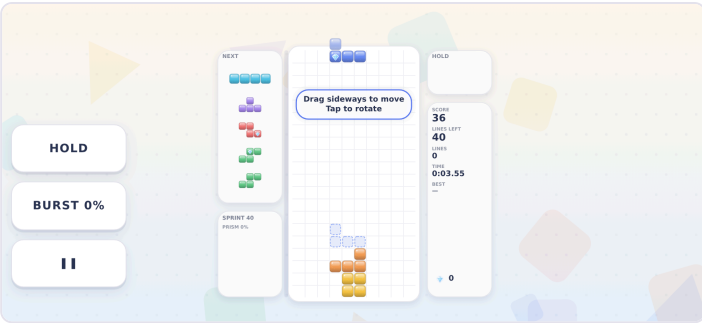
Phone: BURST button glowing when the Prism meter is full



Phone: settings with left-handed layout and vibration toggles



Phone: right-handed layout



Phone landscape (915x412)